Employee Attrition Analysis

Objective

The goal of this project is to analyze employee attrition and identify factors that influence why employees leave a company using Python and Power BI.

Tools Used

Python, Pandas, Matplotlib, Power BI

Dataset

IBM HR Analytics Employee Attrition dataset (Kaggle).

Project Workflow

1. Loaded the dataset using Pandas.
2. Checked data types and missing values.
3. Analyzed employee attrition by department, overtime, job role, and age group.
4. Created visualizations using Matplotlib.
5. Exported the cleaned dataset to CSV.
6. Built an interactive dashboard in Power BI.

Dashboard Components

• Total Employees Card
• Employees Left Card
• Attrition Rate Card
• Attrition by Department
• Attrition by Overtime
• Attrition by Job Role
• Attrition by Age Group
• Interactive slicers for Department, Gender, Job Role, and Overtime

Key Insights

1. Some departments experience higher employee attrition than others.
2. Employees working overtime are more likely to leave the company.
3. Certain job roles show consistently higher attrition levels.
4. Younger employees tend to have a higher risk of attrition.
5. Attrition patterns help HR identify areas requiring improvement.

HR Recommendations

• Reduce excessive overtime.
• Improve employee engagement and work-life balance.
• Offer career development opportunities.
• Focus retention efforts on high-risk departments and job roles.
• Monitor attrition trends regularly using dashboards.

Conclusion

This project demonstrates how Python and Power BI can be combined to analyze HR data, identify employee attrition trends, and provide data-driven recommendations for improving employee retention.